
Dynamic Disequilibrium: An Agent-Based Mechanism for Bubbles without Biased Agents

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Abstract

We present the *Estimated Dynamic Equilibrium Model* (EDEM), an agent-based framework in which supply and demand are realisations of a coupled stochastic process driven by heterogeneous, error-prone agent valuations. The framework’s central technical contribution is a generative mechanism for persistent disequilibrium: when market-clearing prices are repeatedly selected from the upper tail of noisy bid distributions and fed back into future valuations, expected prices drift upward even with strictly zero-mean estimation errors. We derive the bias in closed form for a clean special case ($\sigma(n-1)/(n+1)$ for n iid uniform bids, which is the classical winner’s curse) and show via simulation that compounding the bias across epochs produces exponential price growth without any behavioural assumption about investor optimism. EDEM extends Miller [1977]’s divergence-of-opinion theory to a fully dynamic setting and recovers Walrasian equilibrium and Miller’s static premium as limiting cases. Eight controlled experiments in the Python ABM framework Mesa, on a 32×32 real-estate neighbourhood, plus a 30-cell sensitivity grid, demonstrate six qualitatively distinct regimes—all reachable from the same agent ruleset. The result has direct implications for machine-learning valuation algorithms (Zestimate-style automated valuation models): trained on historical clearing prices, such systems are by construction fitting an upper-order statistic of plausible sale prices rather than fair value, and at deployment time function as positively-biased point estimators.

1 Introduction

The Efficient Market Hypothesis remains the dominant theory of price formation. It predicts when prices reflect available information but does not provide a generative microstructural account of when and why they fail to: bubbles, crashes, persistent cross-sectional return anomalies, and the stylised observation that real markets very rarely sit at the textbook supply–demand fixed point. Decades of empirical work have produced rival theories whose claims often contradict one another, because abundant and noisy financial data can be mined to support nearly any hypothesis [Moffitt, 2017].

The central technical claim of this paper is that two structural features—max-bid clearing and per-epoch feedback of clearing prices into agent valuations—generate persistent positive drift in realised market prices even when individual agents’ estimation errors are exactly zero-mean. The mechanism is order-statistic bias: each seller selects the maximum of multiple noisy bids, and the maximum of n zero-mean draws is, in expectation, strictly above their mean. Feeding the realised maximum back into the next round’s anchor compounds the bias multiplicatively. No behavioural assumption about investor optimism, risk aversion, or attention is required.

35 This is a dynamic generalisation of the classical winner’s curse [Thaler, 1988]: the same upper-tail-
 36 selection effect that biases the expected payment in a single common-value auction, fed epoch-to-
 37 epoch into the next anchor value, produces unbounded drift in market valuations.

38 The implication for machine-learning valuation matters for the NeurIPS audience. An ML valuation
 39 algorithm trained on historical clearing prices—automated valuation models (AVMs), iBuying pricing
 40 models, comparable-sale algorithms—is not fitting fair value. By construction it is fitting the upper-
 41 order statistic of plausible sale prices, because the clearing price is the maximum of the bids received.
 42 At deployment time it functions as a positively-biased point estimator that may itself provide the
 43 market with a coordinating signal pulling realised winning bids further upward.

44 **Contributions.**

- 45 1. An order-statistic mechanism for bubbles without biased agents (Section 4): in closed form, for a
 46 clean special case, the expected winning bid exceeds the home value by $\sigma(n-1)/(n+1)$ when
 47 n bidders draw zero-mean errors with dispersion σ . Compounded across epochs this produces
 48 exponential price drift.
- 49 2. A unified framework (Section 3) that nests Walrasian equilibrium and Miller’s static premium
 50 as limiting cases, and admits a family of out-of-equilibrium regimes (business cycles, persistent
 51 overshoots and undershoots, runaway bubbles, constant transition) parameterised by estimation
 52 noise, balancer strength, and time-on-market.
- 53 3. Eight controlled experiments plus a 30-cell sensitivity grid (Section 6) demonstrating that the order-
 54 statistic drift produces price drift above textbook equilibrium in 100% of parameter combinations
 55 tested.
- 56 4. A discussion of implications for ML valuation systems (Section 7) connecting the model to
 57 selection-aware estimation procedures that AVM deployers should adopt.

58 **2 Background and Related Work**

59 Miller [1977] proposed a static model in which the N shares of a stock are sold to the N most
 60 optimistic of $K > N$ potential investors, so the clearing price reflects the marginal optimist rather
 61 than the mean valuation. Greater divergence of opinion thus produces a price *premium*, and (because
 62 future returns are earned against the elevated price) lower subsequent returns. The half-century of
 63 empirical literature that grew up around this claim has not converged: the *discount* camp argues
 64 divergence of opinion proxies for risk and predicts higher future returns [Williams, 1977, Mayshar,
 65 1983, Varian, 1985, Merton, 1987, Epstein and Wang, 1994], while the *premium* camp finds the
 66 opposite [Doukas et al., 2006]. Moffitt [2017] summarises the impasse.

67 The static framing is the source of the empirical confusion. Miller’s model specifies a one-shot
 68 relationship between beliefs and the market-clearing price. Real markets are continuous processes
 69 in which today’s price feeds into tomorrow’s beliefs. We show that the same upper-tail-selection
 70 mechanism that produces Miller’s premium (and, separately, the winner’s curse [Thaler, 1988] in
 71 common-value auctions) generates a *dynamic* drift when iterated—and that the same underlying
 72 mechanism can produce either a premium or a discount in finite samples, which explains why
 73 empirical work has found both.

74 Closely related work in agent-based economics has used ABMs to study housing markets [Baptista
 75 et al., 2016, Geanakoplos et al., 2012] and behavioural finance generally [Shiller, 2003]. The
 76 contribution here is the structural identification of the selection mechanism, formalised analytically
 77 and validated empirically across a parameter sweep.

78 **3 The EDEM Framework**

79 A market is staged on a finite torus G of homes, with $Q_s(t)$ sellers $\mathcal{S}_t$ (one per home, immobile) and
 80 $Q_d(t)$ buyers $\mathcal{B}_t$ (mobile, walking the torus). Each home h has a fixed unobservable fair value $v^*(h)$
 81 and a market value $v_t(h)$ that evolves. Each agent i valuing home h at tick t draws an estimation

$$E_{i,t}(h) = v_t(h) (1 + \varepsilon_{i,t,h}), \quad \varepsilon_{i,t,h} \stackrel{\text{iid}}{\sim} F_i(\sigma_i), \quad (1)$$

82 with $\mathbb{E}[\varepsilon] = 0$. The three-way index is essential: each *bid* is an independent draw. Throughout, σ_i is
 83 expressed as a decimal (0.05 means $\pm 5\%$ maximum error); parameter tables report percentages.

84 A buyer that lands on a seller's home posts a bid $\beta_{b,s}^{(t)} = E_{b,t}(h(s))$. Sellers accumulate bids; on
 85 patience timeout, the seller offers to sign with the buyer holding the best bid. The buyer commits iff

$$\beta \geq \frac{1}{|\mathcal{B}_b|} \sum_{\beta' \in \mathcal{B}_b} \beta', \quad (2)$$

86 where $\mathcal{B}_b$ is the buyer's outstanding bid set (including β). Eq. (2) is order-statistic selection on the
 87 buyer's side: committing only when the offered bid is at or above the buyer's running benchmark
 88 restricts purchases to the upper tail of the buyer's own bid distribution.

89 The realised market price is the output of a clearing functional $p_t = A(X_t; \theta)$ over the full market
 90 state X_t . We study two instantiations. In the discrete-event *DE* variant, A is the rolling average of
 91 the last W completed sales:

$$p_t = \frac{1}{m_t} \sum_{j=k(t)-m_t+1}^{k(t)} P_j^{\text{sale}}, \quad m_t = \min(W, k(t)), \quad (3)$$

92 where $k(t)$ counts completed sales up to tick t . In the speculative *EDEM* variant there are no realised
 93 sales; instead each home's value updates multiplicatively at the end of every epoch of T ticks:

$$v_{t+T}(h) = v_t(h) \cdot \bar{r}_t, \quad \bar{r}_t = \frac{1}{|\mathcal{Y}_t|} \sum_{s \in \mathcal{W}_b(t)} \min_{b \in \mathcal{Y}_t} \frac{\beta_{b,s}^{(t)}}{v_t(h(s))} \quad (4)$$

94 when the set $\mathcal{Y}_t$ of buyers holding at least one winning bid is non-empty (else $\bar{r}_t = 1$). $\mathcal{W}_b(t)$ is the
 95 set of sellers on which buyer b holds the winning bid.

96 The supply and demand quantities are themselves integer-valued random walks driven by the price
 97 increment $\Delta p_{t-1} = p_{t-1} - p_{t-2}$:

$$Q_s(t) = Q_s(t-1) + Z_t^s, \quad \mathbb{E}[Z_t^s] = C_b \text{sgn}(\Delta p_{t-1}). \quad (5)$$

98 With $C_b > 0$ rising prices add sellers (mean-reverting); with $C_b < 0$ rising prices remove sellers
 99 (trend-following); with $C_b = 0$ the population is frozen. A symmetric equation governs Q_d . A
 100 population floor of one agent per side truncates the random walk in extreme regimes.

101 Setting $\sigma_i \equiv 0$ collapses Eq. (1) to universal agreement; the order-statistic drift vanishes and EDEM
 102 approaches the Walrasian fixed-point equilibrium. Retaining $\sigma_i > 0$ but freezing the dynamics
 103 ($T \rightarrow \infty$, single period, no value update) recovers a Miller-style static premium mechanism.

104 4 The Order-Statistic Mechanism

105 **Closed-form bias.** Consider a seller s receiving $n \geq 2$ bids $\beta_i = v_t(h(s))(1 + \varepsilon_i)$ with $\varepsilon_i \sim$
 106 $\mathcal{U}[-\sigma, +\sigma]$ iid. The expected winning bid is

$$\mathbb{E} \left[\frac{\max_i \beta_i}{v_t(h(s))} \right] = 1 + \mathbb{E} \left[\max_i \varepsilon_i \right] = 1 + \sigma \frac{n-1}{n+1}, \quad (6)$$

107 using the order-statistic identity for the maximum of n iid uniform variates. The right-hand side
 108 strictly exceeds 1 for all $n \geq 2$ and $\sigma > 0$; the gap grows in both n and σ .

109 **Relation to the winner's curse.** Eq. (6) also recovers the classical winner's curse as a static special
 110 case. In a common-value auction, the winning bidder is selected precisely because her signal is the
 111 most favourable draw among the bidder pool. The curse is thus an order-statistic selection effect
 112 before it is a behavioural failure: optimism is not required for the winning bid to be upward-selected.
 113 Rational bid shading can correct the expected payment in a static auction, but it does not remove the
 114 allocation fact that the winning signal comes from the upper tail. EDEM's contribution is to feed that
 115 upper-tail clearing price into the next period's valuation anchor, turning the one-period winner's-curse
 116 effect into a dynamic drift mechanism.

What Eq. (6) does and does not prove. Eq. (6) establishes that the seller-side maximum has expected ratio strictly above 1. The per-epoch update Eq. (4) compounds a related but distinct quantity $\bar{r}_t$ (the buyer-side mean of $\min_s$ winning ratios), for which we have no closed form. Two further care points apply when moving from the theorem to sample-path behaviour: (i) for an iid multiplicative process $v_T = v_0 \prod_t r_t$, the expectation $\mathbb{E}[v_T] = v_0 (\mathbb{E}[r])^T$ grows multiplicatively whenever $\mathbb{E}[r] > 1$, but the median is governed by $\mathbb{E}[\log r]$; by Jensen’s inequality $\mathbb{E}[r] > 1$ does not on its own imply $\mathbb{E}[\log r] > 0$. (ii) The fallback $\bar{r}_t = 1$ for empty-winner epochs puts a point mass at unity that the clean theorem does not capture.

Empirical bridge. Table 1 reports the distribution of $\bar{r}_t$ extracted directly from our EDEM-regime simulations, with $\bar{r}_t = v_{t+T}(h)/v_t(h)$ at every epoch boundary across all seeds. The growth condition $\mathbb{E}[\log \bar{r}_t] > 0$ holds in every regime, certifying sample-path exponential growth and closing the gap between Eq. (6) and the simulated buyer-min update.

Table 1: Empirical distribution of $\bar{r}_t$ across 1500 epochs per regime (10 seeds, 150 epochs/seed). The growth condition $\mathbb{E}[\log \bar{r}_t] > 0$ holds in every regime examined.

Regime	$\mathbb{E}[\bar{r}_t]$	$\Pr(\bar{r}_t > 1)$	$\mathbb{E}[\log \bar{r}_t]$	median $\bar{r}_t$
Run 6 ($C_b = 0, \bar{\sigma} = 0.15$)	1.0429	83.5%	+0.0411	1.044
Run 7 ($C_b = +1, \bar{\sigma} = 0.15$)	1.0104	19.7%	+0.0098	1.000
Run 7 ($C_b = -1, \bar{\sigma} = 0.15$)	1.0153	25.4%	+0.0144	1.000
Run 8 ($C_b = -1, \bar{\sigma}$ growing)	1.0287	24.9%	+0.0236	1.000

5 Implementation

EDEM is implemented in the Python ABM framework Mesa [Masad and Kazil, 2015] on a 32×32 toroidal MultiGrid, matching the wrapped patch world of the original NetLogo prototypes from which this work descends. Per-cell state $(v_t(h), v^*(h))$, last sale price, last sale tick) is a lightweight dataclass attached to each grid cell; buyers and sellers are `mesa.Agent` subclasses. Bids are mirrored dictionaries on each side. The rolling 25-sale market price of Eq. (3) is a `deque(maxlen = 25)`. The per-epoch update in Eq. (4) fires on a cycle-counter trick one tick before buyers reset, preserving their winning-bid flags. The balancer Eq. (5) fires $\lfloor |C_b| \rfloor$ deterministic swaps per epoch plus one Bernoulli swap with probability $|C_b| - \lfloor |C_b| \rfloor$; sign of C_b controls direction. A population floor of one agent per side prevents the balancer from killing the last agent.

The Mesa model and 42 pytest unit tests fit in under 600 lines including docstrings. Each experiment script seeds the model deterministically and runs 10 seeds. Total reproduction time on a 2024-era laptop is approximately 25 minutes (CPU only). Code and data are released under MIT licence at a URL provided in the camera-ready version of this paper.

6 Experiments and Results

Eight controlled experiments grouped into two regimes (DE: Runs 1–5, discrete-event sales; speculative-market EDEM: Runs 6–8, multiplicative value updates) plus a 30-cell sensitivity grid (Run 9). Each experiment runs 10 seeds; figures plot the median across seeds with the 10th–90th percentile band. Full parameter tables are in Appendix B; per-run figures for runs not shown inline here are in Appendix C.

The six regimes. Table 2 consolidates the end-of-run statistics. The eight runs sweep three parameter axes (estimation noise $\bar{\sigma}$, seller patience, balancer C_b) and produce six qualitatively distinct steady-state regimes—band-stable, business-cycle, persistent overshoot, persistent undershoot, runaway bubble, and constant transition—all reachable from the same agent ruleset.

Bubble regime under no balancer. With balancer disabled ($C_b = 0$) and $\bar{\sigma} = 0.15$, the market exhibits a clean log-linear bubble (Fig. 1): the median value-to-true ratio multiplies $\sim 421\times$ over 3000 ticks, with a tight band. The mechanism is exactly that of Section 4: each epoch’s average winning ratio $\bar{r}_t$ is > 1 in expectation, and Eq. (4) compounds it multiplicatively.

Table 2: Eight EDEM runs: parameters and observed outcomes (median across ≥ 8 seeds). DE runs report median deviation of the rolling market price from the textbook equilibrium; EDEM runs report v_t/v^* .

Run	Manipulated parameter	Outcome metric	Regime
1	$\bar{\sigma} = 0.05$, patience ≤ 50	−2% to +11% band	band-stable
2	$\bar{\sigma} = 0.25$	−17% to +34% periodic	business-cycle
3	patience ≤ 100	+14% persistent	overshoot
4	demand intercept −50	−18% persistent	undershoot
5	shock at $t = 3000$, then re-shocks	moving-target chase	transitional
6	EDEM, $C_b = 0$, $\bar{\sigma} = 0.15$	$v_t/v^* = 421\times$	runaway bubble
7	EDEM, $C_b \in \{-1, 0, +1\}$	7.7, 421, 4.2 $\times$ resp.	balancer-shaped
8	EDEM, $C_b = -1$, $\bar{\sigma}$ growing	$v_t/v^* = 30\times$	super-linear

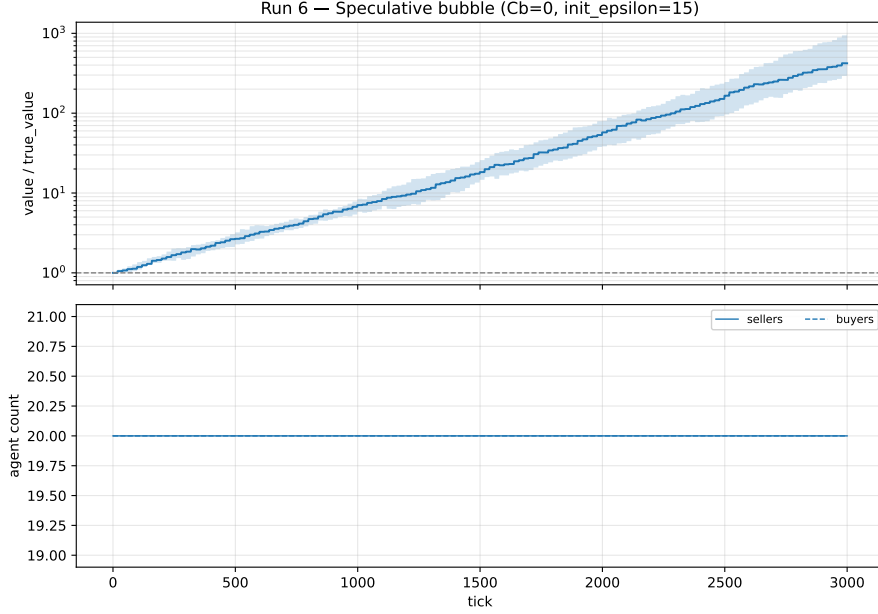


Figure 1: Run 6 (bubble, $C_b = 0$, $\bar{\sigma} = 0.15$). Top: median v_t/v^* (log scale) with 10–90 percentile band. Bottom: agent counts (frozen at 20 each because $C_b = 0$).

157 **Balancer-coefficient sweep.** Varying $C_b \in \{+1, 0, -1\}$ at fixed $\bar{\sigma} = 0.15$ yields three qualitatively
158 distinct trajectories (Fig. 2): the mean-reverting case ($C_b = +1$) plateaus near $4\times$ as the population
159 drifts toward sellers; the trend-following case ($C_b = -1$) reaches $\sim 7.7\times$ with the population drifting
160 toward buyers; the no-balancer case dominates both at $\sim 421\times$. This refines a prior claim in the
161 literature that trend-following should be the most explosive regime: in the Mesa simulation, the
162 trend-following balancer drains sellers to a single agent, leaving fewer total sellers to inflate; the
163 no-balancer case maintains balanced populations and admits all of the upward bid-selection bias
164 unfiltered. No setting of C_b in this grid restores $v_t/v^* = 1$.

165 **Sensitivity to C_b and $\bar{\sigma}$.** A 5×6 grid sweeps $C_b \in \{-1, -0.5, 0, +0.5, +1\}$ and $\bar{\sigma} \in$
166 $\{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$, with 5 seeds per cell at 1500 ticks (Fig. 3). Every cell of the grid
167 has $v_t/v^* > 1$, and 43% of cells exceed $10\times$ within 1500 ticks. The order-statistic drift is universal
168 across the parameter grid, not a corner case. $C_b = 0$ dominates within every $\bar{\sigma}$ column.

169 **Other regimes.** DE Runs 1–5 produce the band-stable, business-cycle, overshoot, undershoot, and
170 transitional regimes listed in Table 2. Run 4 (low agent density): mean realised price $p \approx 41$ versus
171 textbook $p^* = 50$, an 18% shortfall, with episodes reaching $p \approx 37$ (26% at the trough). Run 5

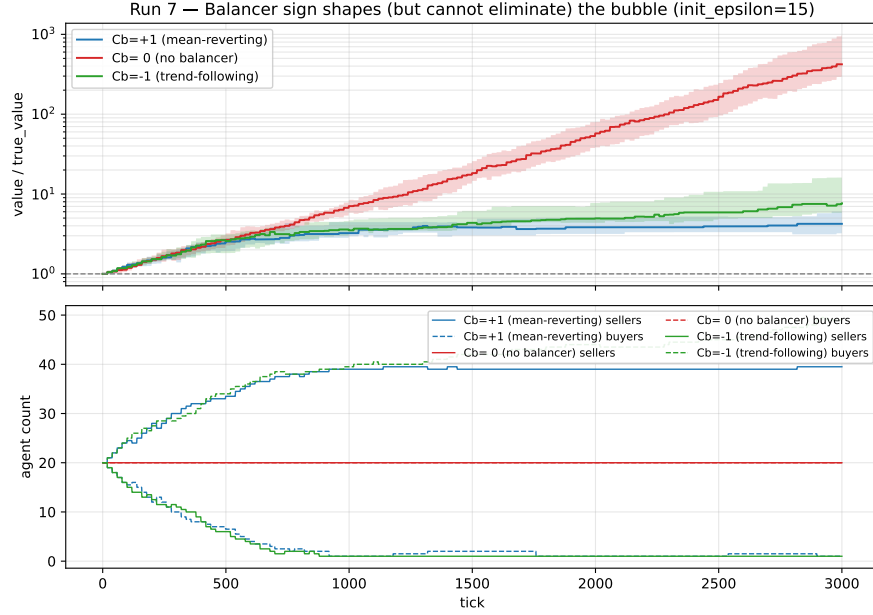


Figure 2: Run 7 (balancer sweep at $\bar{\sigma} = 0.15$). The three regimes are overlaid; $C_b = 0$ produces the strongest bubble. The population panel shows the one-vs-many state the strong balancers induce.

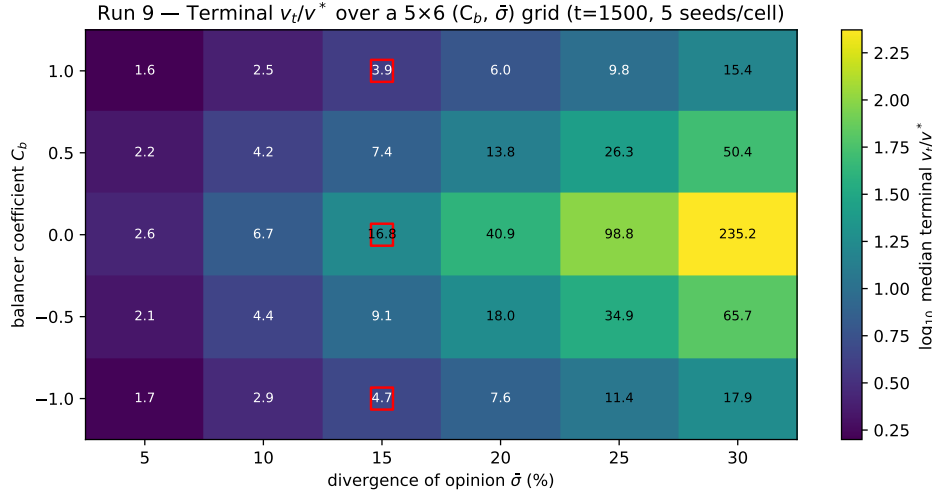


Figure 3: Run 9 (sensitivity heatmap). Median terminal v_t/v^* over a 5×6 (C_b , $\bar{\sigma}$) grid, log colour scale. Red squares mark cells corresponding to Runs 6 and 7. Every cell exceeds 1.0; 43% exceed 10.0.

172 (shocked market): the realised price never settles when demand is toggled every 2000 ticks. The
 173 figures for these runs are in Appendix C.

174 7 Discussion: Implications for ML Valuation

175 **Bid-selection asymmetry, not estimation bias, drives bubbles.** The estimation function Eq. (1)
 176 has zero mean error: an agent's expected valuation is exactly the home's current value. Yet the
 177 EDEM speculative regime produces unbounded upward drift even with this unbiased estimator. The
 178 mechanism is not behavioural bias but a statistical asymmetry: each seller selects the maximum bid

received, and the maximum of N noisy estimates of a fair value is, in expectation, larger than the fair value itself. Eq. (4) compounds this asymmetry multiplicatively across epochs. This is the dynamic generalisation of the classical winner’s curse [Thaler, 1988].

Implication for ML valuation systems. The same mechanism operates in any pricing system whose inputs are anchored to recent market-clearing prices. A machine-learning valuation algorithm trained on historical sale prices is, by construction, fitting the upper-order statistic of plausible sale prices for a similar asset—the maximum of the relevant noisy estimates—not their mean. Deployed at scale, such an algorithm functions as a positively-biased point estimator that may provide the market with a coordinating signal pulling realised winning bids further upward, feeding back into the next training cycle.

A defensible deployment would correct for this: predict the median rather than the mean of plausible sale prices; use a censored-data likelihood that accounts for the offer-acceptance threshold; or use an explicitly rank-aware estimator that targets fair value rather than winning-bid value. The 2021 wind-down of Zillow’s algorithmic-iBuying programme [Zillow Group, Inc., 2021] is one episode in which a clearing-price-anchored valuation system at scale incurred substantial losses; we do not advance EDEM as a complete account of that specific event, but the public record is consistent with the mechanism here.

Why empirical premium/discount studies disagree. Each empirical sample of the divergence-of-opinion premium is a finite-window observation along a coupled stochastic process governed by σ and C_b . Two studies sampling distinct positions on distinct trajectories may both correctly observe opposite signs. The unifying claim from EDEM is not that one camp is right and the other wrong, but that the static framing has been asking the wrong question; the relevant quantity is the trajectory of $\bar{r}_t$, not its sign at a snapshot.

8 Limitations and Conclusion

EDEM models a single asset class with no leverage, no macroeconomic environment, and non-adaptive (non-learning) agents. The 30-cell sensitivity grid covers a defensible range of $(C_b, \bar{\sigma})$ but does not explore extreme $|C_b|$ values where the population dynamics qualitatively change. Run 5’s shock-recovery scenario uses a 12,000-tick budget that is too short for the market to reach its post-shock floor before the patience boost fires. Bayesian-belief and reinforcement-learning extensions of the agent rule set (Appendix D) are sketched but not implemented.

The contribution we believe travels beyond the EDEM setting is the mechanism. Bubbles, persistent disequilibrium, and the empirical contradictions in the divergence-of-opinion literature do not require behavioural bias to explain. Repeated upper-tail selection, fed back into subsequent valuations, is sufficient. For the NeurIPS audience the practical consequence is concrete: an ML valuation model trained on historical clearing prices is by construction fitting an upper-order statistic, and selection-aware estimation is a deployment-time requirement, not an academic nicety.

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A ODD Protocol

This appendix documents EDEM in the standard ODD protocol for agent-based models [Grimm et al., 2020].

Purpose and patterns. EDEM characterises the conditions under which an agent-based real-estate market reaches a stable equilibrium price, and identifies the mechanisms that prevent equilibrium when they do not. Patterns the model reproduces: stable band-bounded equilibria; endogenous business cycles; persistent shifts in the realised price relative to the textbook equilibrium under altered patience or density; multiplicative price drift in the absence of a balancer.

Entities, state variables, scales. Entities: Buyers and Sellers (`mesa.Agent` subclasses), Homes (per-cell dataclass on a 32×32 toroidal grid). Buyer state: epsilon (estimation-error bound), delay, current bid, heading; EDEM variant adds the lowest-bid-to-value ratio and a yellow (winning) flag. Seller state: epsilon, patience timer, current ask price, dictionary of received bids; EDEM adds number of bids, total bids, best-bid-to-value, best-bid-to-true-value. Home state: market price, last-sale tick, last-sale price, fair value v^* , current value v . Model state: rolling window of last 25 sale prices (DE), epoch counter (EDEM), cross-population epsilon (EDEM). One spatial unit = one home; one tick is the smallest temporal unit. DE balance period is 100 ticks; EDEM epoch is 20.

Process overview. Per tick: (i) each agent steps once, in randomised order over both classes; a Buyer’s step posts bids, then wiggles, then steps forward; a Seller’s step processes patience and best-bid logic; (ii) the model invokes the balancer (DE: every 100 ticks; EDEM: every 20 ticks via the cycle-counter trick that preserves buyers’ winning-bid flags); (iii) the data collector samples model-level reporters.

Design concepts. *Basic principles:* EDEM extends the divergence-of-opinion premise of Miller [1977] to a multi-period setting; agents act on noisy estimates of an unobservable fair value. *Emergence:* the price level, agent population dynamics, and macroscopic regimes emerge from local interaction; none of these phenomena are coded in directly. *Adaptation:* sellers lower ask prices when patience elapses without high-enough bids (the only adaptive behaviour). Buyers do not adapt. *Objectives:* sellers maximise realised sale price subject to patience. Buyers commit only to offers whose realised bid is at or above their own running benchmark (Eq. (2))—a selection rule, not a price-minimisation objective. *Learning, prediction:* none in the baseline model. *Sensing:* buyers sense whether a seller is on their patch; sellers sense the bids in their own bid table; neither senses the global market price directly. *Interaction:* direct via mirrored-dictionary updates; indirect via the average ask price and rolling sale-price average influencing newly-spawned agents. *Stochasticity:* agent placement, buyer heading, epsilon draws, per-bid error draws, victim selection, Bernoulli draws for fractional C_b swaps—all from a single seeded `numpy.random.Generator`. *Observation:* per-tick model reporters stacked into Parquet datasets, one per seed.

Initialisation. DE: spawn `equi_qnty` sellers at random unique cells with patience $\sim \mathcal{U}[0, \text{init_patience})$ and ask price drawn from $\mathcal{U}[-\bar{\sigma}, +\bar{\sigma}]$ around p^* ; spawn `equi_qnty` buyers at random cells. EDEM: 20 sellers and 20 buyers, all homes initialised with $v_0(h) = v^*(h) = 100$.

Input data and submodels. No external input data; EDEM is closed. Estimation function: Eq. (1). Bid acceptance: Eq. (2). Market price: Eq. (3) (DE) and Eq. (4) (EDEM). Balancer: Eq. (5) in finite-population form with population floor of one.

B Parameter Tables

Common parameters across all runs. World size 32×32 toroidal; seeds per run ≥ 8 (typically 10); bid-acceptance rule `netlogo` (the rule actually coded in the original 2018 NetLogo source, matching Eq. (2)).

DE runs 1–5. Supply $\hat{Q}_s(p) = a_s + 0.5p$, $a_s = 0$. Demand $\hat{Q}_d(p) = a_d - 0.5p$, $a_d = 100$ (Runs 1–3, 5) or 50 (Run 4) or $\{125, 75\}$ alternating (Run 5 Scenario B). Maximum valuation error $\bar{\sigma}$: 0.05

307 (Runs 1, 3–5) or 0.25 (Run 2). Maximum patience: 50 (Runs 1, 2, 4) or 100 (Run 3) or $50 \rightarrow 165$
 308 at $t = 7000$ (Run 5 Scenario A). Balance period $T_B = 100$. Sale window $W = 25$. Ticks per seed:
 309 20,000 (Runs 1–4), 12,000 (Run 5).

310 **EDEM runs 6–8.** Initial $\bar{\sigma}$: 0.15 (Runs 6, 7) or 0.05 (Run 8). $\bar{\sigma}$ growth per epoch: 0 (Runs 6, 7) or
 311 $+0.005$ (Run 8). Balancer C_b : 0 (Run 6); $\{+1, 0, -1\}$ (Run 7); -1 (Run 8). Initial buyers / sellers:
 312 20/20. Epoch length 20 ticks. Ticks per seed: 3000. True value $v^*(h) = 100$ uniform across all
 313 homes.

314 **Run 9 sensitivity grid.** $C_b \in \{-1, -0.5, 0, +0.5, +1\}$; $\bar{\sigma} \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$. 5
 315 seeds per cell, 1500 ticks each.

316 C All Experiment Figures

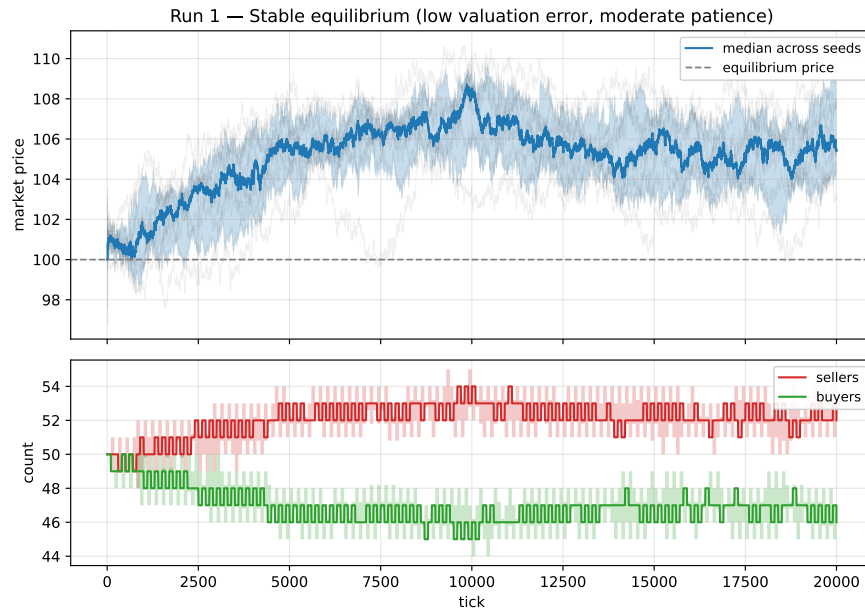


Figure 4: Run 1 – Stable equilibrium with low valuation error and moderate patience.

317 D Extensions and Future Work

318 The framework of Section 3 is deliberately minimal so that extensions can be plugged in without
 319 rewriting the agent skeleton. Five directions are especially fruitful and we sketch each below.

320 **Bias-corrected clearing.** The bid-selection asymmetry embedded in the max-bid clearing rule
 321 cannot be cancelled by any linear balancer within the explored grid. Two replacements for the clearing
 322 functional A in Eq. (3) would address it directly: (i) a winsorised mean replacing $\bar{r}_t$ in Eq. (4); (ii) a
 323 selection-aware estimator substituting the median or lower-percentile of all bids (not only winning
 324 bids) for $\bar{r}_t$. Either is a one-line edit to the reference implementation.

325 **Realtors and commission structures.** The DE seller is a strict utility-maximiser whose only
 326 friction is patience. Real-world residential transactions are intermediated by realtors whose payoff is
 327 a fraction of the sale price. A realtor extension would replace the seller agent with an aggregated
 328 listing agent whose patience and ask-price strategies optimise expected commission across a portfolio.
 329 This is a natural setting for empirical calibration: realtor commission rates and listing durations are
 330 publicly observable, unlike the patience parameter of the bare DE seller.

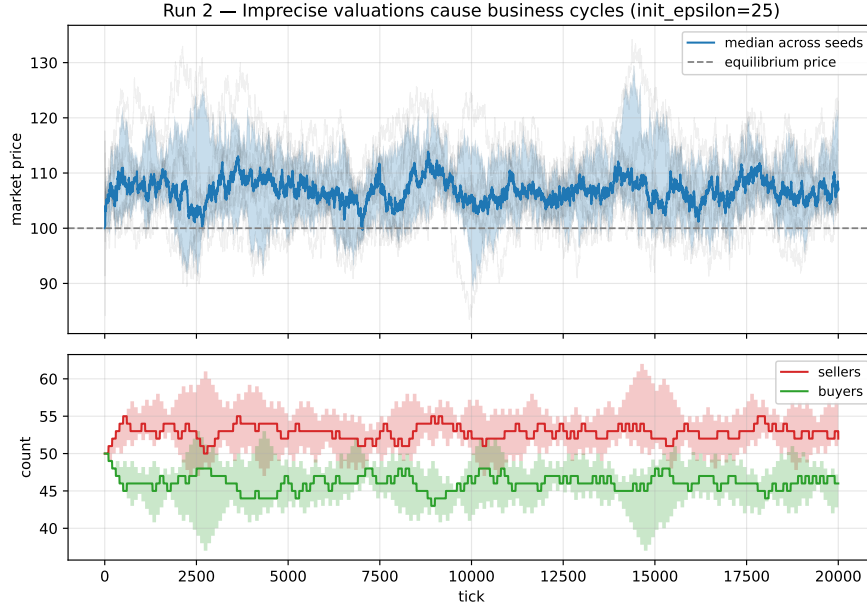


Figure 5: Run 2 – High valuation error ($\bar{\sigma} = 0.25$) generates endogenous business cycles.

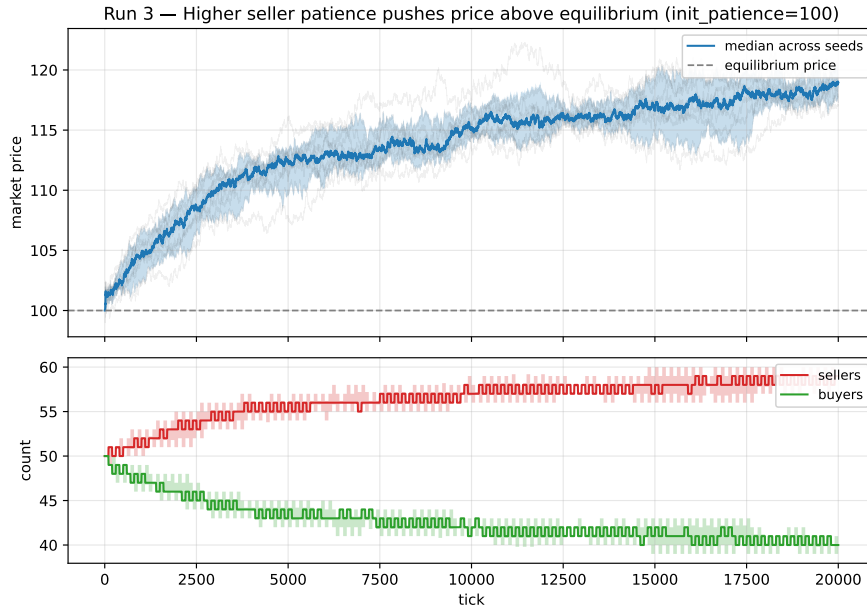


Figure 6: Run 3 – Doubling seller patience to 100 pushes the market price about 14% above the textbook equilibrium and sustains it there.

331 **Open vs. blind auctions.** EDEM as posed is a blind-bid model. Open-bid auctions substantially
 332 change the bid distribution by exposing buyers to one another's valuations and would likely amplify
 333 the bid-selection asymmetry.

334 **Construction and the supply side.** A construction-market variant inverts EDEM's agent roles:
 335 contractors bid downward to win projects, with the lowest-bid contractor winning. The selection
 336 asymmetry then operates in reverse, biasing realised contract prices below the population fair value.

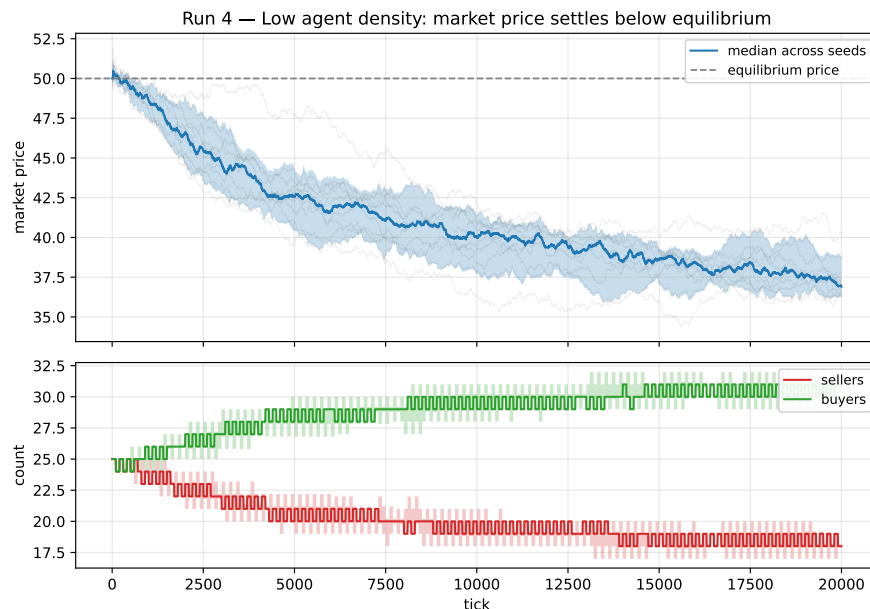


Figure 7: Run 4 – Halving the demand intercept reduces equilibrium quantity to 25; the realised market price settles about 18% below the textbook equilibrium.

Adaptive agents. EDEM agents are non-learning. A Bayesian extension where each agent maintains a posterior over fair value and updates after every observed sale would converge, in the limit, toward bias-corrected estimation. A reinforcement-learning variant with a buy-low/sell-high reward would discover and exploit the bid-selection asymmetry; this would be a computational analogue to the Zillow-Offers iBuying problem [Zillow Group, Inc., 2021] and would offer a stress test of any proposed clearing-rule fix.

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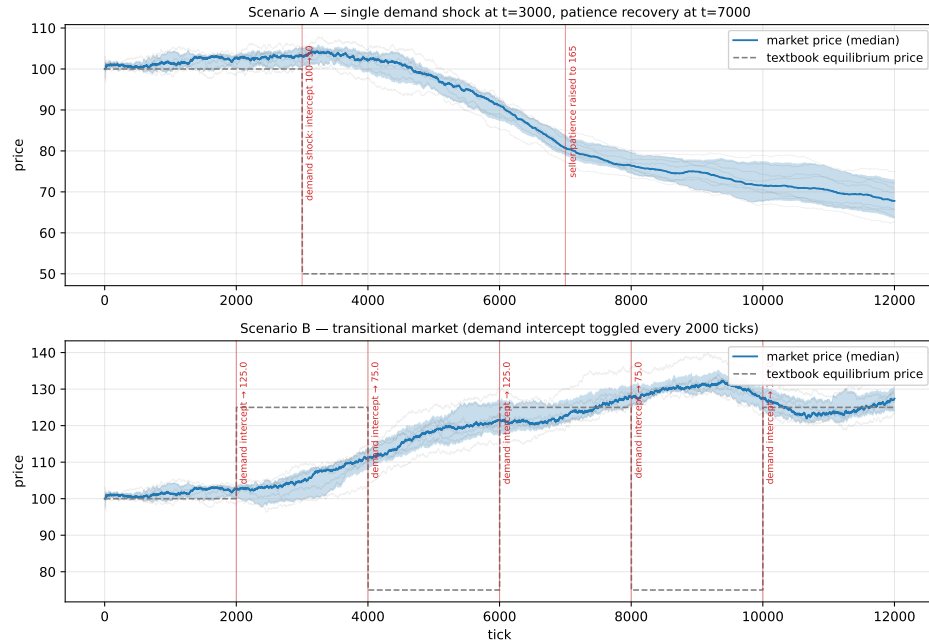


Figure 8: Run 5 – Two shock scenarios. Top: a single demand shock at $t = 3000$ followed by a patience boost at $t = 7000$. Bottom: a transitional market in which demand toggles every 2000 ticks.

Answer: [Yes]

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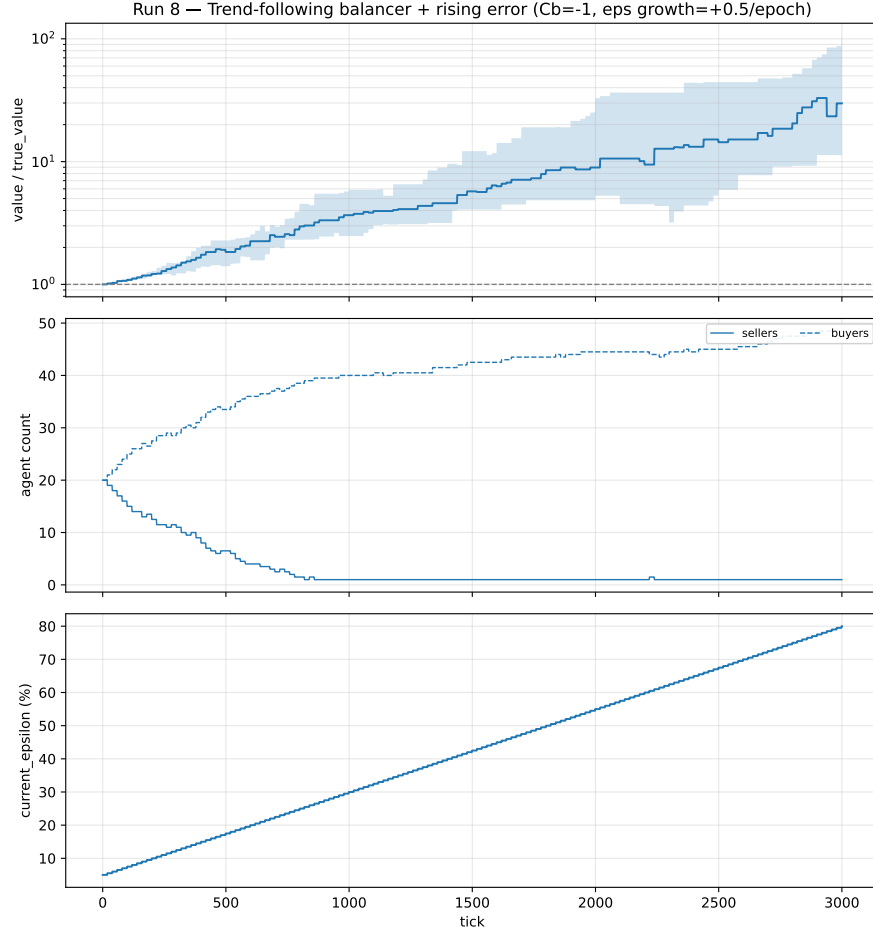


Figure 9: Run 8 – Trend-following balancer ($C_b = -1$) combined with linearly-growing $\bar{\sigma}$ ($0.05 \rightarrow 0.80$). The trajectory is sustained super-linear rather than visibly double-exponential.

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